

Intelligent Mobile Robot Platform

smBaseD

1. Product Overview

smBaseD, our intelligent mobile robot platform is built to serve the needs of our robots, forming the core structure for all robotic solutions in the Smaït product line. Designed for durability, adaptability, and precision, this chassis is integral to the development, scalability, and deployment of Smaït technologies across various applications. Featuring advanced mobility systems, modular compatibility, and robust materials, smBaseD seamlessly integrates cutting-edge technologies, including multi-sensor fusion, SLAM-based positioning and mapping, multiple operational modes, and coordinated multi-robot operation. Enhanced by cloud-based management, IoT control systems, automated charging, and flexible motion control, it ensures exceptional adaptability, precision, and efficiency while minimizing operational complexity and costs in dynamic environments.

2. Product Schematic



3. Product Specifications

3.1 Basic Information

Parameter		Specs
Basic Parameters	Dimensions	500mmL * 500mmW * 310mmH
	Weight	35kg
	Max Load	59kg
	No-load Battery Life	12h
	Charging Time	4h
	Maximum Load Capacity	50kg
	Battery Capacity	24V/20AH = 480Wh
	Chassis Form	2* Drive Wheels + 4* Omni-directional + 1* Fixed Wheel
	Main Hardware Configuration	6.5-inch High Load Solid Wheels (Lightweight Type) *2
		RGBD Camera Sensor (0.6-8m) *1
Ultrasonic Distance Sensors (0.3-1.2m) *1		
IMU Six-axis Sensor *1		
	High-performance X86 Industrial Computer *1	
Motion Parameters	Maximum Navigation Speed	0.8m/s (Adjustable Speed)
	Obstacle Avoidance Distance	80cm
	Chassis Inclination	5°
	Obstacle Clearance Height	10mm
	Minimum Obstacle Clearance	40mm
	Turning Radius	0mm
	Maximum Rotation Speed	60°/s
	Positioning Accuracy	±5cm
	Mapping Area	10000m²(Up to 30000m²)
	Repeat Positioning Accuracy	±5cm
Expansion Interface	Power Input Interface	24V (200W Power)
	USB Ports	2 Ports, Expandable
	Network Interface	RJ45, Optional Wireless Network
	Physical Expansion Interface	Structural Accessories
	Other Interfaces	System Switch, Thermal Switch
Charging Parameters	Input Voltage	AC 110-240V
	Output Voltage	29.4V
	Maximum Charging Current	7A
	Protection Mechanism	Short Circuit, Overcurrent, etc.
Environmental Requirements	Storage Temperature	Temperature: -10°C~60°C
	Operating Temperature	Temperature: 5°C~40°C

3.2 API Port Interface

No.	Function	Description
1	Network Connection	Web socket server connection/disconnection.
2	Robot Position	Coordinates of the robot on the map.
3	Full Status Information	Full status (battery level, navigation status, software/hardware emergency stop, floor name).
4	Complete Map Data	Display map.
5	Radar Data	Display radar data.
6	Navigation Path Planning	Display the planned navigation path.
7	Navigation Status	The current navigation status returned.
8	Detailed Recharge Status	Return status information for recharging.
9	Speed Control	Manual remote control of the robot.
10	Publish Target Point (Coordinate Mode)	Select a point on the map for navigation.
11	Cancel Navigation	Cancel current navigation.
12	Set Initial Position (Coordinate Mode)	Select a point on the map to set as the initial position.
13	Set Recharge Dock Position	Select a point on the map to set as the recharge dock position.
14	Soft Emergency Stop	Enable/disable soft emergency stop.
15	Virtual Wall Setting (Overlay Mode)	Set virtual wall information in one go.
16	Virtual Wall Setting (Append Mode)	Append virtual wall information to system cache.
17	Mark Point (Overlay Mode)	Set mark point information in one go.
18	Mark Point (Append Mode)	Append mark point information to system cache.
19	Insert Current Position as Mark Point	Set the current robot position as a mark point.
20	Map Editing Function	Interface for map editing functionality.
21	Recharge Dock Alignment	Start recharge alignment independently (infrared alignment only, no navigation).
22	Program Control Framework	Control overall robot state, switch between mapping, navigation, and map modes.
23	List System Map	List all available maps in the current system.
24	Delete Map	Delete a specified map and its information.
25	Get Current Mark Point List	Retrieve the list of mark points associated with the current map.
26	Delete All Mark Points	Delete all mark points associated with the current map.
27	Delete Single Mark Point (By Name)	Delete a specific mark point by name.
28	Get Current Virtual Wall List	Retrieve the list of virtual walls associated with the current map.
29	Delete All Virtual Walls	Delete all virtual walls associated with the current map.
30	Delete Single Virtual Wall (By Name)	Delete a specific virtual wall by name.
31	Publish Single Navigation	Navigate to a specific point by its name.

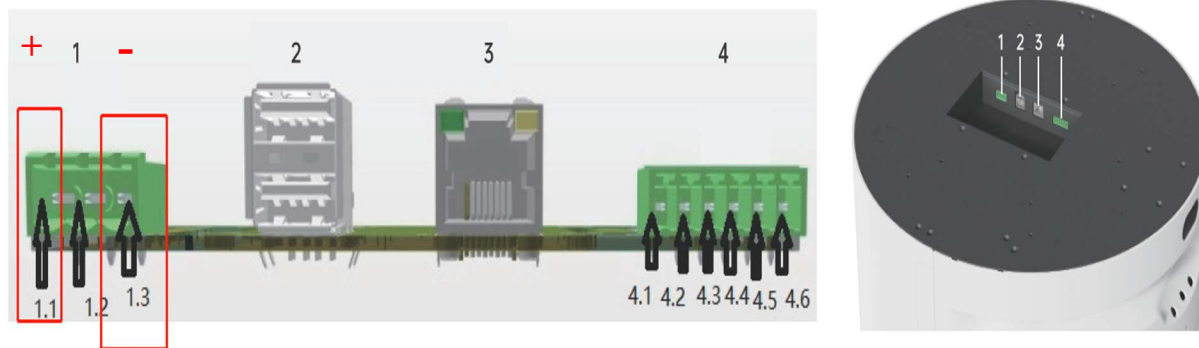
	(By Name)	
32	Set Initial Position (By Name)	Set the initial position by its name.
33	Multi-Point Navigation Control (By Name)	Start or stop multi-point navigation.
34	Recharge Control Interface	Start or stop recharging.
35	Get Current Map Metadata	Retrieve the map's coordinate origin and dimensions.
36	Modify Navigation Speed	Adjust the robot's movement speed during navigation.
37	Version Update	Update code via USB or network.

3.3 Product Function

No.	Function	Description
1	Mapping	Uses SLAM algorithm and sensor data for incremental mapping in unknown areas, with loop closure functionality.
2	High-Precision Localization	Performs high-precision localization using sensor data in known maps, effectively correcting the robot's position in cases of slight slippage.
3	Path Planning	Plans a reasonable and efficient path for the robot to follow after setting a target point.
4	Autonomous Obstacle Avoidance	Avoids unknown obstacles during navigation based on the global path derived from path planning.
5	Autonomous Charging	Returns to the charging dock when the battery is low or on user command, using infrared sensor data to locate and dock autonomously.
6	Multi-Point Navigation	Executes tasks with multi-point navigation programs on the robot, useful for offline operations.
7	Ultrasonic-Assisted Obstacle Avoidance	Optional feature to enable ultrasonic-assisted obstacle detection for detecting special obstacles like glass.
8	Floor Map Management	Saves and loads maps by building floor names for easy switching between floor maps.
9	Map Management	Saves mapping information, loads maps for localization and navigation, and switches between multiple maps.
10	Map Editing	Edits created maps to optimize display, mark additional obstacles, or clear unnecessary ones.
11	Wireless Remote Control	Controls the robot's movement via a mobile app.
12	Virtual Wall	Allows users to set virtual walls to restrict the robot from entering specific areas during autonomous navigation.
13	Upper-Level Communication	Enables interaction with the robot via network communication and supports cross-platform development.
14	Depth Camera Stereo Obstacle Avoidance	Uses a depth camera to detect obstacles in a three-dimensional space, complementing non-lidar-level obstacle information.

15	Software/Hardware Emergency Stop	Supports emergency stop via software or hardware.
16	Manual Relocation	Allows manual setting of the robot's position on the map when starting navigation away from the charging dock.
17	Remote/Local Code Updates	Supports algorithm and code updates via network or USB insertion.
18	Dynamic Navigation Speed Adjustment	Enables setting of different speed modes for robot navigation.
19	Multi-Attribute Mark Point Management	Supports setting attributes for marked points (e.g., recharge points), triggering automatic charging processes when navigating to those points.
20	Global Self-Localization	Automatically calibrates the robot's position on the map using sensor data or assists manual relocation calibration.

3.4 Product External Connection Interface



1: Power Port: 24V/10A with overcurrent and short circuit protection, and current monitoring.

- 1.1: Positive
- 1.3: Negative

2: USB Expansion Interface: USB 2.0 expansion port x2.

3: Network Port: Used for chassis communication, TCP/IP protocol.

- Upper Port: Upper Computer Camera
- Lower Port: External USB

4: Switch Interface: Used to connect emergency stop devices.

- 4.2 and 4.3 ports are for emergency stop
- 4.5 and 4.6 are for system startup

PIN Pitch Connector: 5.08mm and 3.81mm